TRUST GAP

IN LLM APIs

Specified model

Served model

Verifiable?

Are You Getting What You Pay For?

Auditing Model Substitution in LLM APIs: Why Software Tests Fail and Hardware Attestation Wins

arxiv.org/abs/2504.04715

Commercial APIs may covertly serve cheaper models — provably verifiable only with hardware attestation.

The Model Substitution Problem

Economic incentives drive covert swaps in black-box LLM APIs

Formal Framing: Hypothesis Testing

H_0: API serves P_spec(y|x) — the advertised model distribution
H_1: API serves P_actual(y|x) — a substituted model distribution

Four Substitution Types

1 Smaller Model

e.g. 8B instead of 70B

2 Quantized Variant

FP8 / INT8 instead of FP16

3 Fine-Tuned Version

Updated or customised weights

Why Providers Substitute

- GPU costs dominate inference spend
- Black-box API hides served model
- FP8 cuts memory & cost ~2×
- Benchmark scores nearly identical
- No cryptographic proof of model identity

The core challenge: no ground truth. API responses alone cannot prove model identity.

FP8 Quantization Preserves Most Accuracy — Making Substitution Hard to Detect

Benchmark scores for original vs. FP8-quantized models across five model families

Model	MMLU	GSM8K	MATH	GPQA Diamond
Llama-3-8B-Instruct	62.69 ± 0.18	61.14 ± 3.47	20.65 ± 3.43	22.62 ± 0.22
└ FP8	62.43 ± 0.26	60.90 ± 4.10	14.91 ± 2.49	20.14 ± 0.24
Llama-3-70B-Instruct	78.05 ± 0.08	88.06 ± 1.44	35.69 ± 1.33	29.60 ± 0.51
└ FP8	77.88 ± 0.13	87.35 ± 1.24	35.75 ± 1.16	33.12 ± 0.30
Gemma-2-9b-it	71.86 ± 0.08	81.80 ± 1.35	33.41 ± 0.28	28.64 ± 2.97
└ FP8	71.92 ± 0.11	79.41 ± 1.14	32.53 ± 0.34	27.81 ± 3.22
Qwen2-72B-Instruct	82.18 ± 0.08	86.72 ± 1.00	37.39 ± 1.41	29.93 ± 2.71
└ FP8	81.98 ± 0.08	86.82 ± 0.97	37.67 ± 1.39	31.08 ± 1.96
Mistral-7B-Instruct-v0.3	59.15 ± 0.10	35.90 ± 4.54	8.94 ± 1.24	21.60 ± 0.17
└ FP8	58.77 ± 0.13	32.20 ± 4.02	7.68 ± 1.23	22.72 ± 0.19

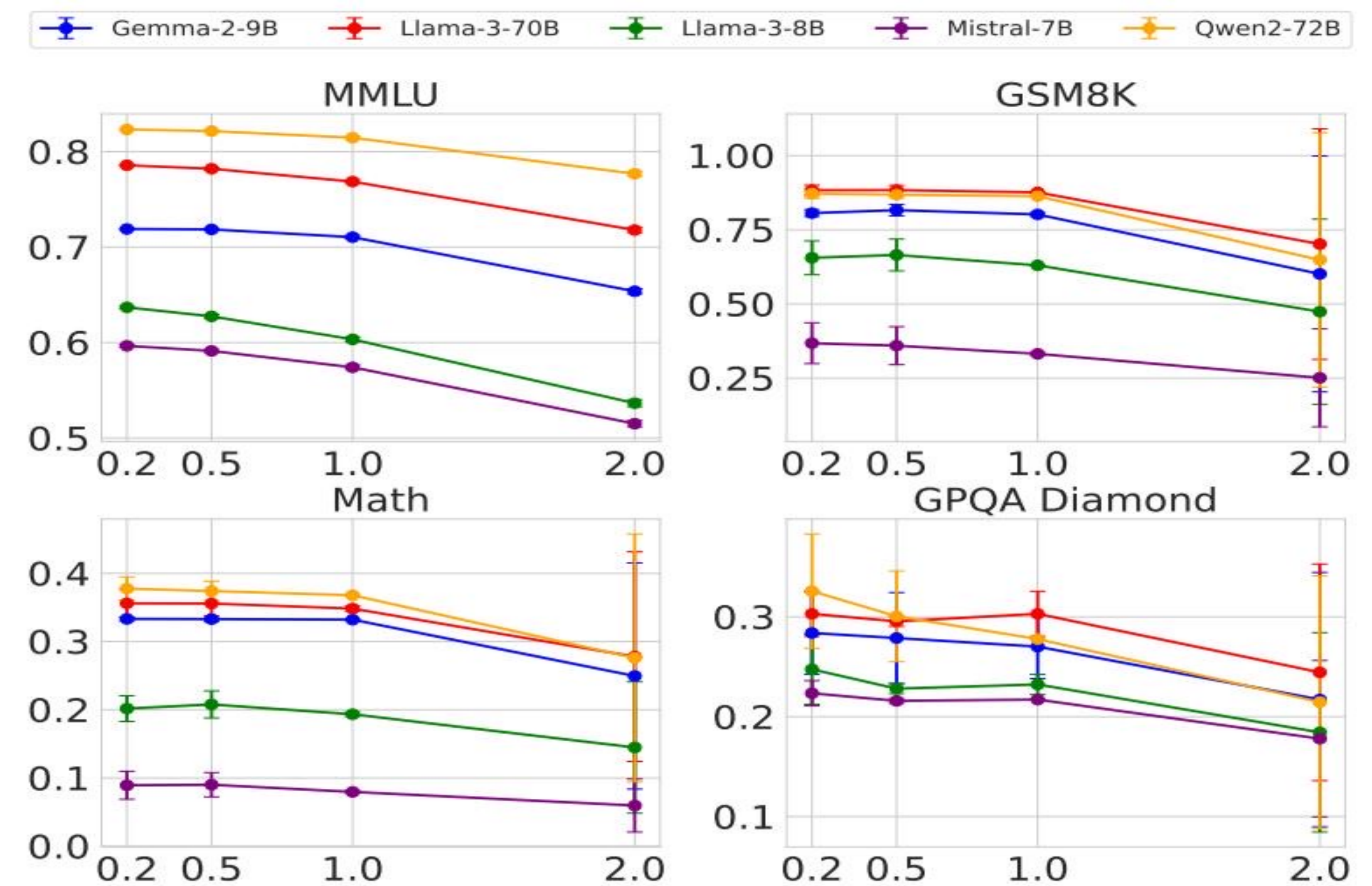
Key Finding

- Most FP8 scores fall within the original's error bars.
- MMLU and GSM8K show virtually no degradation.
- MATH shows the largest drops — still within noise.
- Indistinguishable by benchmark alone.

FP8 quantization ≈ 2× cost reduction with near-zero benchmark penalty → ideal covert substitute

Higher Temperature Amplifies Gaps but Increases Noise

Benchmark accuracy vs. temperature — figure_1 from the paper



Interpretation

Low temp (0.2)

Performance stable; original and FP8 nearly overlap.

Mid temp (1.0)

Slight divergence visible but within statistical noise.

High temp (2.0)

All models degrade sharply — gap becomes noise-dominated.

Implication

Temperature cannot be used as a reliable discriminator between original and quantized model.

Source: Figure 1 — Benchmark accuracy across temperatures 0.2, 0.5, 1.0, 2.0 for five model families.

Log Probability Fingerprinting Fails Due to Inference Nondeterminism

Same model, different frameworks and hardware → indistinguishable log-prob variation

Model	BERT Acc	T5 Acc	GPT2 Acc	LLM2Vec Acc
Llama3-70B-Instruct-FP8	50.55	50.10	50.45	49.90
Llama3-70B-Instruct-INT8	51.60	49.90	51.30	50.25
Gemma2-9b-it-FP8	49.95	50.30	51.20	49.80
Gemma2-9b-it-INT8	49.00	49.70	51.65	49.55
Mistral-7b-v3-Instruct-FP8	50.55	49.75	48.70	48.75
Mistral-7b-v3-Instruct-INT8	49.70	50.75	50.50	51.15
Qwen2-72B-Instruct-FP8	50.05	50.25	48.75	49.80
Qwen2-72B-Instruct-INT8	50.75	50.55	49.45	50.20

Why This Defeats Detection

Framework gap

transformers 4.40 vs 4.50
vLLM 0.6.2 vs 0.8.2
→ different log-probs for same model

Hardware gap

A100 vs H100 produce
different numerical results
for identical model weights

Net result

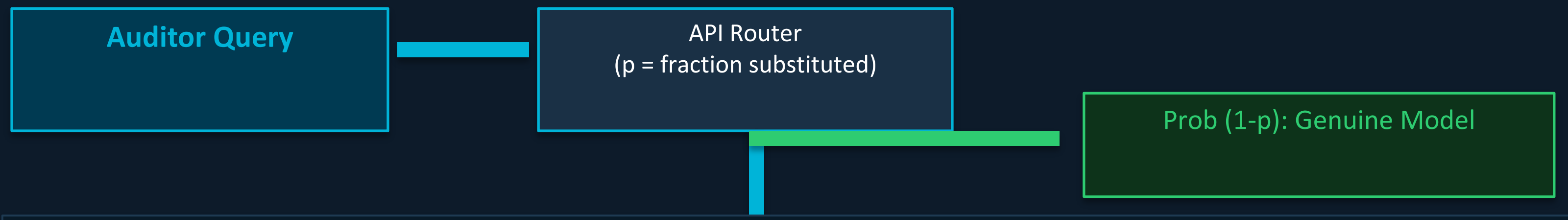
Auditor cannot distinguish
legitimate infra variation
from actual substitution

Source: Figure 5 — Log-probability variation across inference environments for Llama-3-8B.

Randomized Routing at Low Rates Makes Statistical Detection Query-Prohibitive

Providers mix substitute responses with probability p — as p→0, detection requires exponentially more queries

Adversarial Randomized Substitution Model



Mixed Distribution: $P\_mix = (1-p) \cdot P\_spec + p \cdot P\_sub$

As $p \to 0$: $P\_mix \to P\_spec \implies$ test power $\to 0 \implies$ queries required $\to \infty$

Queries Needed to Achieve 95% Power (illustrative)

Substitution Rate (p)	Required Queries (est.)	Practical?
p = 1.0 (full swap)	~100–500	Marginal
p = 0.5 (50% swap)	~1,000–5,000	Expensive
p = 0.1 (10% swap)	~50,000+	Prohibitive
p = 0.01 (1% swap)	~5,000,000+	Impossible

Benchmark Evasion Attack

Provider fingerprints known benchmark prompts and routes them to the genuine model — all other traffic gets the substitute.

Effect on Detection

- Benchmark audits show 100% correct model
- Real-world traffic served by substitute
- Standard audits are systematically blind
- Requires zero-day prompts to detect

Software-Only Auditing Methods Are Fundamentally Unreliable

Comparison matrix of three failure modes

Method Type	Core Weakness	Practical Limitation
Text-Based Statistical Tests (e.g. MMD on outputs)	Need statistically large query volumes; subtle substitutions produce tiny distributional shifts	Impractically large query budgets (thousands to millions); fails against quantized models or low-p randomised routing
Log-Probability Fingerprinting	Production inference is nondeterministic across software versions (transformers, vLLM) and GPU hardware (A100/H100)	Legitimate infra variation is statistically indistinguishable from model substitution; false positive rate is uncontrollable
Adversarial Counter-measures (Benchmark Evasion + Randomised Routing)	Active adversary adapts: routes benchmark prompts to genuine model; mixes substitute at low rate p	All passive software tests defeated simultaneously; detection requires unrealistic query volumes or unpublished benchmark prompts

Verdict: No software-only audit method achieves reliable detection under realistic adversarial conditions.

Trusted Execution Environments Provide Provable Cryptographic Model Integrity

Hardware attestation: the only currently viable path to verifiable LLM inference

TEE Architecture

- Hardware Root of Trust
- ⊂ Attestation Certificate (Remote)
- ⊂ ⊂ Sealed Model Weights
- ⊂ ⊂ ⊂ Verified Inference Code
- ⊂ ⊂ ⊂ ⊂ Encrypted I/O Channel

TEE vs. Software-Only: Property Comparison

Property	Software Audit	TEE Attestation
Model identity proof	✗ None	✓ Cryptographic
Weights verification	✗ Impossible	✓ Hash-sealed
Inference code check	✗ Black-box	✓ Verified
Overhead	Low	Modest (~10-15%)
Adversary-proof	✗ Bypassable	✓ Hardware-bound
Deployable today	✓ Yes	✓ Yes (H100 CC)

- NVIDIA Confidential Computing on H100 GPUs provides a production-ready TEE deployment path today.
- Unlike Zero-Knowledge Proofs (computationally prohibitive at scale), TEEs add only modest overhead (~10–15%).
- Remote attestation lets any auditor cryptographically verify model weights and code without trusting the provider.

Key Takeaways: Close the Verification Gap with Hardware Attestation

Evidence-based findings and a call to action for the research community

01 Model Substitution Is Economically Rational and Practically Undetectable via Software

FP8 quantization cuts GPU costs ~2× while preserving near-identical benchmark scores. Black-box APIs provide no ground truth for model identity.

02 FP8 Quantization Is Near-Invisible in Benchmarks

Across five model families (Llama, Gemma, Qwen, Mistral) and four benchmarks, FP8 scores overlap original error bars on MMLU and GSM8K — detection is statistically intractable.

03 Inference Nondeterminism Defeats Log-Probability Fingerprinting

Software frameworks (transformers, vLLM) and GPU hardware (A100 vs H100) introduce log-prob variation that is indistinguishable from substitution — even for the correct model.

04 Randomized Substitution and Benchmark Evasion Defeat Statistical Tests

At $p = 1\%$, detecting substitution requires millions of queries. Benchmark evasion routes audit queries to the genuine model while regular traffic is served by the substitute.

05 Trusted Execution Environments Are the Only Viable Path to Verifiable Inference

TEEs provide cryptographic hardware attestation of model weights and inference code with modest overhead (~10–15%). NVIDIA H100 Confidential Computing is deployable today.

Call to Action: Standardise TEE-based model attestation in commercial LLM APIs — require providers to publish verifiable hardware attestation reports.